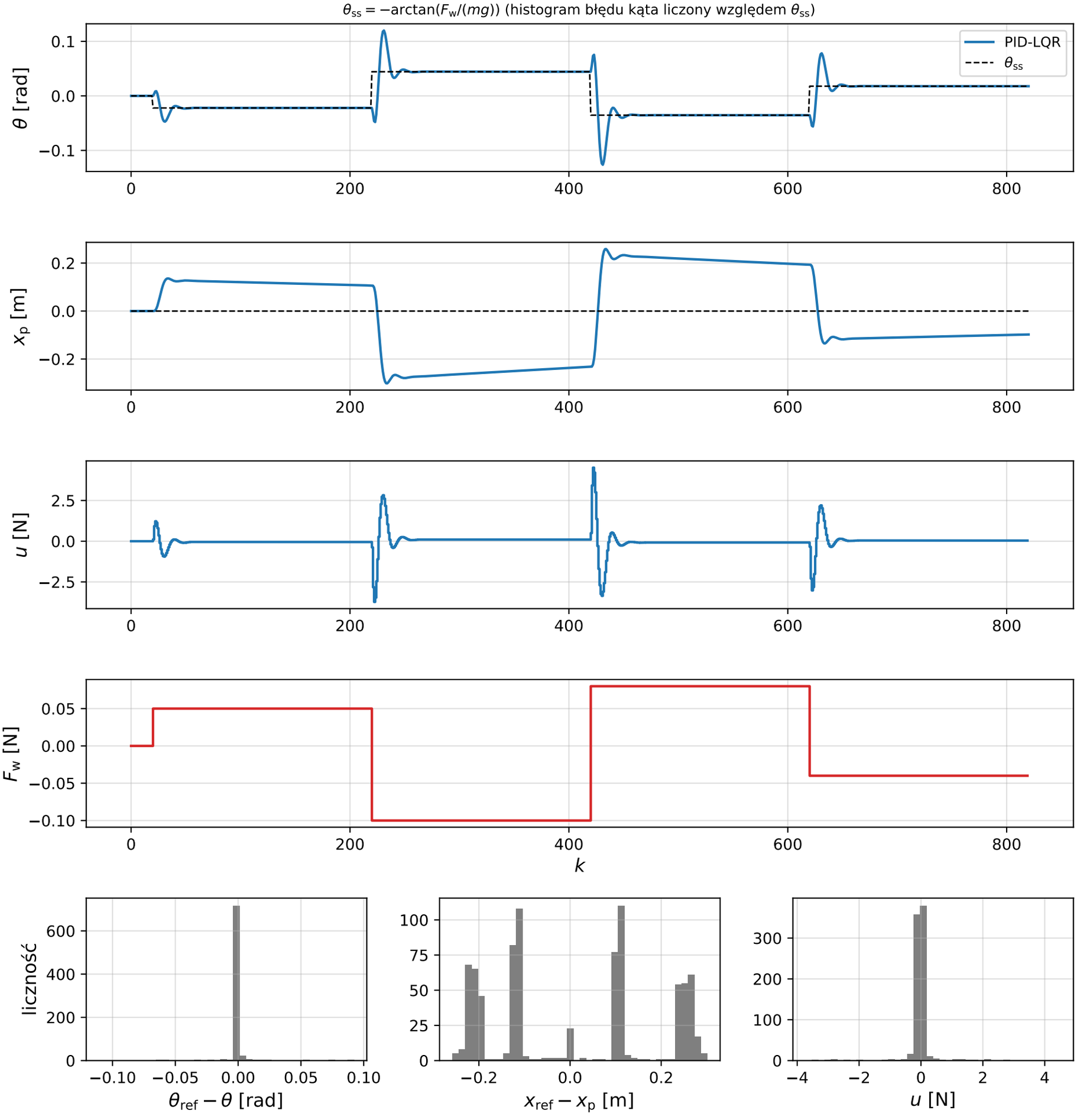
$\theta_{ss} = -\arctan(F_w/(mg))$ (histogram błędu kąta liczony względem θ_{ss})
PID-LQR
θ_{ss}
θ [rad]
x_p [m]
u [N]
F_w [N]
k
liczność
$\theta_{ref} - \theta$ [rad]
$x_{ref} - x_p$ [m]
u [N]